

MODULE - 4

Numerical Methods

Errors in Numerical Computation:

Numerical methods generally gives approximate solution to a given mathematical problem. The approximate solution obtained will involve an error which is precisely the difference between the exact solution and approximate solution.

Mainly there are three types of errors that may occur in most of the numerical methods. They are round-off error, ^{we measure} truncation error and ^{also} relative error.

Round-off Errors occur because of the computing device's inability to deal with certain numbers.

Ex: $\pi = 3.1415926535 \dots$. Such numbers need to be rounded off to some near approximation which is dependent on the word size used to represent numbers of the device.

Truncation Error refers to an error in a method, which occurs because some series of steps is truncated (shortened) to a fewer number. For example, we approximate the e^x by the 1st four term of its Maclaurin series as, in $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$

$$\text{Truncation Error} = e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!}\right) = \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

$$\text{Relative Error} = \frac{\text{True value} - \text{approximate value}}{\text{True value}}.$$

Solutions of Equations:

Numerical methods used for solving equations is based on Intermediate value theorem which is as follows. "If $f(x)$ is continuous in $[a, b]$ and if $f(a)$ and $f(b)$ are of opposite signs, then the equation $f(x)=0$ will have at least 1 real root between a and b ."

Newton-Raphson Method:

- * Consider the given equation in the form $f(x)=0$
- * Find the values 'a' and 'b' such that $f(a)$ and $f(b)$ are of opposite signs.
- * Choose $x_0 = a$ or b whichever is more closer to the exact root. (The value of x_0 for which $f(x_0)$ is closer to zero).
- * Using the iterative formula,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, \dots$$

we can find the root of $f(x)=0$ to the desired accuracy.

Q Find the root of $2x^3 - 3x = 6$ using Newton-Raphson method correct to 5 decimal places.

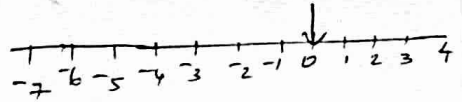
Sol:

$$f(x) = 2x^3 - 3x - 6$$

$$f(0) = -6$$

$$f(1) = -7 \text{ (+ve)}$$

$$f(2) = 4 \text{ (-ve)}$$



The required root lies between 1 and 2.

We choose $x_0 = 2$, since corresponding to 2, $f(2) = 4$ is more closer to 0 than $f(1) = -7$.

Now, Iterative formula is,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{--- ①}$$

$$\begin{aligned} \text{put } n=0 \text{ in ① ; } x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= 2 - \frac{f(2)}{f'(2)} \\ &= 2 - \frac{4}{21} \\ &= 1.809523 \end{aligned} \quad \left\{ \begin{array}{l} f(x) = 2x^3 - 3x - 6 \\ f'(x) = 6x^2 - 3 \\ x_0 = 2 \end{array} \right. \quad \left\{ \begin{array}{l} f(2) = 2 \cdot 2^3 - 3 \cdot 2 - 6 \\ \quad = 16 - 6 - 6 = 4 \\ f'(2) = 6 \cdot 2^2 - 3 \\ \quad = 21 \end{array} \right.$$

$$\begin{aligned} \text{put } n=1 \text{ in ① ; } x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 1.809523 - \frac{f(1.809523)}{f'(1.809523)} \\ &= 1.78420 \end{aligned}$$

$$\text{Put } n=2 \text{ in } \textcircled{1} ; x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 1.783769$$

$$\text{Put } n=3 \text{ in } \textcircled{1} ; x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$= 1.783769061$$

$\therefore$ Approximate root is 1.78376

Q Find the approximate root of the equation $e^x \sin x = x^2$ correct to 3 decimal places using Newton Raphson method. (Initial value: $x_0 = 3$)

Sol: $f(x) = e^x \sin x - x^2$; $f'(x) = e^x \cos x + \sin x \cdot e^x - 2x$
 $= e^x [\cos x + \sin x] - 2x$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{--- } \textcircled{1}$$

put $n=0$ in $\textcircled{1}$; $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ (Given: $x_0 = 3$)

$$= 3 - \frac{f(3)}{f'(3)}$$

$$= 3 - 0.2675$$

$$= 2.7325$$

put $n=1$ in $\textcircled{1}$; $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$$= 2.7325 - \frac{f(2.7325)}{f'(2.7325)}$$

$$= 2.632$$

$$\begin{aligned} \text{put } n=2 \text{ in } \textcircled{1}, \quad x_3 &= x_2 - \frac{f(x_2)}{f'(x_2)} \\ &= 2.632 - \frac{f(2.632)}{f'(2.632)} \\ &= 2.6182 \end{aligned}$$

$$\begin{aligned} \text{put } n=3 \text{ in } \textcircled{1}, \quad x_4 &= x_3 - \frac{f(x_3)}{f'(x_3)} \\ &= 2.6182 - \frac{f(2.6182)}{f'(2.6182)} \\ &= 2.618 \end{aligned}$$

$\therefore$ Approximate root is 2.618

[9] Find the square root of 28 correct to 3 decimal place using Newton Raphson method.

sol: Let $x = \sqrt{28} \Rightarrow x^2 = 28 \Rightarrow x^2 - 28 = 0$

Here: $f(x) = x^2 - 28$

$f(5) = -3$ (-ve)

$f(6) = 8$ (+ve)

Root lies between 5 and 6. Choose $x_0 = 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{--- } \textcircled{1} \quad \boxed{f'(x) = 2x}$$

put $n=0$ in $\textcircled{1} \Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 5 - \frac{f(5)}{f'(5)} = 5.3$

put $n=1$ in $\textcircled{1} \Rightarrow x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 5.3 - \frac{f(5.3)}{f'(5.3)} = 5.291509$

put $n=2$ in $\textcircled{1} \Rightarrow x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 5.2915026$

$\therefore$ square root of 28 is 5.2915 approximately. i.e. $\sqrt{28} \approx 5.2915$
 $\approx \underline{\underline{5.292}}$

Another Method

[Q] Find the root of the equation $x^2 + x - 1 = 0$ correct to 3 decimal places using Newton's method with $x_0 = 0$

sol: $f(x) = x^2 + x - 1$

$$x_0 = 0$$

$$f'(x) = 2x + 1$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\Rightarrow x_{n+1} = x_n - \frac{x_n^2 + x_n - 1}{2x_n + 1}$$

$$\begin{cases} f(x) = x^2 + x - 1 \\ f(x_n) = x_n^2 + x_n - 1 \end{cases}$$

$$\begin{cases} f'(x) = 2x + 1 \\ f'(x_n) = 2x_n + 1 \end{cases}$$

$$= \frac{(2x_{n+1})x_n - (x_n^2 + x_n - 1)}{2x_n + 1}$$

$$= \frac{2x_n^2 + x_n - x_n^2 - x_n + 1}{2x_n + 1} = \frac{x_n^2 + 1}{2x_n + 1}$$

$$\text{ie } x_{n+1} = \frac{x_n^2 + 1}{2x_n + 1} \quad \text{--- (1)}$$

$$\text{put } n=0 ; \quad x_1 = \frac{x_0^2 + 1}{2x_0 + 1} = 1$$

$$\text{put } n=1 ; \quad x_2 = \frac{x_1^2 + 1}{2x_1 + 1} = 0.667$$

$$\text{put } n=2 ; \quad x_3 = 0.619$$

$$\text{put } n=3 ; \quad x_4 = 0.618$$

$$\text{put } n=4 ; \quad x_5 = 0.618$$

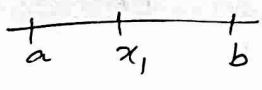
$\therefore$ The required root is 0.618 approximately

$$\begin{cases} \text{Given} \\ x_0 = 0 \end{cases}$$

Regula-Falsi Method:

Consider the equ. $f(x)=0$. Let a, b ($a < b$) be the two values of x such that $f(a)$ and $f(b)$ are of opposite signs. Then, 1st approximation to the root is

$$x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)} \quad \text{--- ①}$$

If $f(x_1)$ and $f(a)$ are of opposite sign,  then the root lies b/w 'a' and 'x₁'.

So, we replace 'b' by x_1 in ①, and we get second approximation x_2 .

If $f(x_1)$ and $f(a)$ are of same sign, then $f(x_1)$ and $f(b)$ will be of opposite sign. Therefore the root lies between x_1 and b . Hence we replace 'a' by ' x_1 ' in ① and get 2nd approximation x_2 .

The process is to be repeated till the root is found to have desired accuracy.

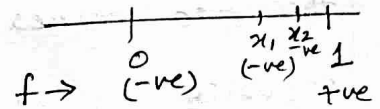
Note: Regula-Falsi method is known as the 'Method of False position'.

Q Determine the root of $xe^x - 2 = 0$ by Regula-Falsi method.

Sol. Let $f(x) = xe^x - 2$

$$f(0) = -2 \text{ (-ve)}$$

$$f(1) = 0.718 \text{ (+ve)}$$



$\therefore$ Root lies between 0 and 1.

$$\boxed{a=0, b=1}$$

1st approximate root is,

$$x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

$$= 0.735759$$

$$\begin{cases} f(x) = xe^x - 2 \\ f(a) = f(0) = -2 \\ f(b) = f(1) = e - 2 \end{cases}$$

$$f(x_1) = -0.464 \text{ (-ve)}$$

$\therefore$ Root lies between x_1 and 1.

$$\text{Choose } a = x_1 = 0.735759$$

$$b = 1$$

2nd approximate root is,

$$x_2 = \frac{x_1 f(1) - 1 \cdot f(x_1)}{f(1) - f(x_1)} = 0.8394881$$

$$f(x_2) = -0.056 \text{ (-ve)}$$

$\therefore$ Root lies between x_2 and 1.

$$\text{Choose } a = x_2 = 0.8394881$$

$$b = 1$$

$$3^{\text{rd}} \approx \text{to root is } x_3 = \frac{x_2 f(1) - 1 \cdot f(x_2)}{f(1) - f(x_2)} = 0.851101$$

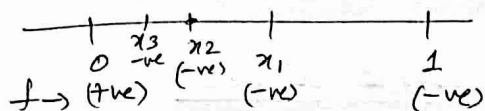
$$\text{Similarly, } 4^{\text{th}} \approx \text{to root is } x_4 = 0.85244 //$$

[9] Perform four iterations of regula falsi method to obtain the true root of $x^3 - 5x + 1 = 0$

Sol: $f(x) = x^3 - 5x + 1$

$f(0) = 1$ (+ve)

$f(1) = -3$ (-ve)



$\therefore$ Root lies between 0 and 1. $a = 0, b = 1$

1st $\approx$ make root : $x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)}$ { $\begin{cases} f(x) = x^3 - 5x + 1 \\ f(0) = 1 \\ f(1) = -3 \end{cases}$

$= 0.25$

$f(x_1) = -0.234$ (-ve)

$\therefore$ Root lies between 0 and x_1 .

Choose : $a = 0, b = x_1 = 0.25$

2nd $\approx$ make root : $x_2 = \frac{af(x_1) - x_1f(a)}{f(x_1) - f(a)} = 0.202593$

$f(x_2) = -4.6497 \times 10^{-3}$

$\therefore$ Root lies between 0 and x_2 .

Choose : $a = 0, b = x_2$

3rd $\approx$ make root : $x_3 = \frac{af(x_2) - x_2f(a)}{f(x_2) - f(a)} = 0.2016553$

$f(x_3) = -7.64365 \times 10^{-5}$

$\therefore$ Root lies between 0 and x_3 .

Choose : $a = 0, b = x_3$

$$4^{\text{th}} \approx \text{mate root : } x_4 = \frac{a f(x_3) - x_3 f(a)}{f(x_3) - f(a)}$$

$$= \underline{\underline{-2016399}}$$

Home works

- [Q] Using regula-Falsi method, compute the real root of the equation $x^3 - 4x - 9 = 0$.
- [Q] Apply Newton Raphson method to evaluate approximately $\sqrt{12}$.
- [Q] (KTU) [7 marks] Use Newton-Raphson method to find a non-zero solution of $x = 2 \sin x$. Start with $x_0 = 1$

Q] Derive an iteration formula for $\sqrt[3]{N}$ where N is a real number and hence find $\sqrt[3]{41}$ correct to four decimal places using Newton Raphson method.

sol: Let $x = \sqrt[3]{N} \Rightarrow x^3 = N \Rightarrow x^3 - N = 0$

$$f(x) = x^3 - N = 0$$

By Newton Raphson method, iteration formula is,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\Rightarrow x_{n+1} = x_n - \frac{(x_n^3 - N)}{3x_n^2}$$

$$= \frac{3x_n^3 - x_n^3 + N}{3x_n^2} = \frac{2x_n^3 + N}{3x_n^2}$$

$$\begin{cases} f(x) = x^3 - N \\ f(x_n) = x_n^3 - N \\ f'(x_n) = 3x_n^2 \end{cases}$$

Thus, iteration formula for $\sqrt[3]{N}$ is,

$$x_{n+1} = \frac{2x_n^3 + N}{3x_n^2} \quad \text{--- (A)}$$

To find $\sqrt[3]{41}$: $x = \sqrt[3]{41}$
 $\Rightarrow x^3 = 41$

$$f(x) = x^3 - 41$$

$f(0) = -ve$, $f(1) = -ve$, $f(2) = -ve$, $f(3) = -14$, $f(4) = +23$

$\therefore$ Root lies between 3 and 4. Choose $x_0 = 3$

Now, $x_{n+1} = \frac{2x_n^3 + 41}{3x_n^2}$ --- (1) from (A), putting $N = 41$

put $n=0$ in (1) $\rightarrow x_1 = \frac{2x_0^3 + 41}{3x_0^2} = \frac{2 \times 3^3 + 41}{3 \times 3^2} = 1.17284$

put $n=1$ in (1) $\rightarrow x_2 = \frac{2x_1^3 + 41}{3x_1^2} =$

put $n=2$ in (1) $\rightarrow x_3 = \frac{2x_2^3 + 41}{3x_2^2} =$

put $n=3$ in (1) $\rightarrow x_4 =$

Q Find an iterative formula to find the reciprocal of a number 'N' and hence find the value of $\frac{1}{19}$.

SA: $x = \frac{1}{N} \Rightarrow x - \frac{1}{N} = 0 \Rightarrow f(x) = x - \frac{1}{N} \Rightarrow f(x_n) = x_n - \frac{1}{N}$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f'(x) = 1 \text{ (not possible).}$$

$f'(x_n)$ must involve a variable

So, we take, $N = \frac{1}{x} \Rightarrow f(x) = N - \frac{1}{x}$ $x = \frac{1}{N} \Rightarrow N = \frac{1}{x}$

$$\therefore f(x_n) = N - \frac{1}{x_n}$$

$$f'(x_n) = 0 - \left(-\frac{1}{x_n^2}\right) = \frac{1}{x_n^2}$$

To form an iterative formula, we start with;

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{(N - \frac{1}{x_n})}{\frac{1}{x_n^2}} = \left(x_n \cdot \frac{1}{x_n^2} - N + \frac{1}{x_n}\right) \div \frac{1}{x_n^2}$$

$$= \left(\frac{1}{x_n} - N + \frac{1}{x_n}\right) \times x_n^2 = \left(\frac{2}{x_n} - N\right) x_n^2 = 2x_n - Nx_n^2$$

Thus, $x_{n+1} = 2x_n - Nx_n^2$ is the iterative formula for $\frac{1}{N}$.

To find $\frac{1}{19}$:

$$x = \frac{1}{19} \Rightarrow x - \frac{1}{19} = 0 \Rightarrow f(x) = x - \frac{1}{19}$$

Root lies between 0 and 1.

$$f(0) = -\frac{1}{19} = -0.052 \text{ (-ve)}$$

Choose $x_0 = 0$

$$f(1) = 1 - \frac{1}{19} = 0.88 \text{ (+ve)}$$

Iteration formula is,

$$x_{n+1} = 2x_n - Nx_n^2. \text{ Here, } N = 19 \therefore$$

$$x_{n+1} = 2x_n - 19x_n^2 \text{ --- (1)}$$

$$n=0 \rightarrow x_1 = 2x_0 - 19x_0^2 = 0$$

So, we choose ^{new}₁ $x_0 = 0.02$

$$\text{put } n=0 \text{ in (1), } x_1 = 2x_0 - 19x_0^2 = 2(0.02) - 19(0.02)^2 \\ = 0.0324$$

$$\text{put } n=1 \text{ in (1), } x_2 = 2x_1 - 19x_1^2 \\ = 2(0.0324) - 19(0.0324)^2 \\ = 0.0448$$

$$\text{put } n=2 \text{ in (1), } x_3 = 0.0514$$

$$\text{put } n=3 \text{ in (1), } x_4 = 0.0526$$

$$\text{put } n=4 \text{ in (1), } x_5 = 0.0526$$

Thus, ~~the~~ solution is $x = 0.0526$ correct to four decimal place.

Finite Differences:

Let $y = f(x)$ be a function and ' Δx ' denote the increment in ' x '. Then the 1st finite difference of the function y is defined as,

$$\Delta y = \Delta f(x) = f(x + \Delta x) - f(x).$$

Finite differences of higher orders can be found out using the general formula,

$$\Delta^n y = \Delta(\Delta^{n-1} y); \quad n = 2, 3, 4, \dots$$

Forward differences:

Let x take the values x_0, x_1, x_2, x_3, x_4 and $y = f(x)$ take the values, y_0, y_1, y_2, y_3, y_4 respectively. Then first forward difference is denoted by Δy_k and is defined as

$$\Delta y_k = y_{k+1} - y_k. \text{ Thus, } \Delta y_0 = y_1 - y_0, \Delta y_1 = y_2 - y_1, \text{ etc.}$$

We can illustrate the higher order forward differences using the table below:

Forward difference table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0				
x_1	y_1	$y_1 - y_0 = \Delta y_0$			
x_2	y_2	$y_2 - y_1 = \Delta y_1$	$\Delta y_1 - \Delta y_0 = \Delta^2 y_0$		
x_3	y_3	$y_3 - y_2 = \Delta y_2$	$\Delta y_2 - \Delta y_1 = \Delta^2 y_1$	$\Delta^2 y_1 - \Delta^2 y_0 = \Delta^3 y_0$	
x_4	y_4	$y_4 - y_3 = \Delta y_3$	$\Delta y_3 - \Delta y_2 = \Delta^2 y_2$	$\Delta^2 y_2 - \Delta^2 y_1 = \Delta^3 y_1$	$\Delta^3 y_1 - \Delta^3 y_0 = \Delta^4 y_0$

Backward Differences:

Let x takes the values x_0, x_1, x_2, x_3, x_4 and $y = f(x)$ takes the values, y_0, y_1, y_2, y_3, y_4 respectively. Then first backward difference is denoted by ∇y_k and is defined as, $\nabla y_k = y_k - y_{k-1}$. Thus,

$\nabla y_1 = y_1 - y_0$, $\nabla y_2 = y_2 - y_1$, $\nabla y_3 = y_3 - y_2$ - - - etc. We can illustrate the higher order backward differences using the table below:

Backward difference table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0				
x_1	y_1	$y_1 - y_0 = \nabla y_1$			
x_2	y_2	$y_2 - y_1 = \nabla y_2$	$\nabla y_2 - \nabla y_1 = \nabla^2 y_2$		
x_3	y_3	$y_3 - y_2 = \nabla y_3$	$\nabla y_3 - \nabla y_2 = \nabla^2 y_3$	$\nabla^2 y_3 - \nabla^2 y_2 = \nabla^3 y_3$	
x_4	y_4	$y_4 - y_3 = \nabla y_4$	$\nabla y_4 - \nabla y_3 = \nabla^2 y_4$	$\nabla^2 y_4 - \nabla^2 y_3 = \nabla^3 y_4$	$\nabla^3 y_4 - \nabla^3 y_3 = \nabla^4 y_4$

KDU-VQ2

- Q Find all the first and second order forward and backward differences of y for the following set of (x, y) values:

$(0.5, 1.13)$, $(0.6, 1.19)$, $(0.7, 1.26)$, $(0.8, 1.34)$

Sol:

x	y	1 st order differences	2 nd order differences	
0.5	$1.13 = y_0$			
		0.06		
0.6	$1.19 = y_1$		0.01	
		0.07		0
0.7	$1.26 = y_2$		0.01	
		0.08		
0.8	$1.34 = y_3$			

we have $\Delta y_k = y_{k+1} - y_k$

and $\nabla y_k = y_k - y_{k-1}$

1st order forward differences are,
 $\Delta y_0 = 0.06, \Delta y_1 = 0.07, \Delta y_2 = 0.08$

2nd order forward differences are,

$\Delta^2 y_0 = 0.01, \Delta^2 y_1 = 0.01$

$\Delta y_0 = y_1 - y_0$

But

$y_1 - y_0 = \Delta y_1$

1st order backward differences are,

$\nabla y_1 = 0.06, \nabla y_2 = 0.07, \nabla y_3 = 0.08$

2nd order backward differences are

$\nabla^2 y_2 = 0.01, \nabla^2 y_3 = 0.01$

Interpolation:

If the values of $f(x)$ are known for $x = x_0, x_1, \dots, x_n$ then the process of finding the value of $f(x)$ for any other value of x in the interval (x_0, x_n) is called interpolation and the process of finding the value of $f(x)$ for a value of x outside the interval (x_0, x_n) is called extrapolation.

Newton-Gregory Forward Interpolation Formula: (EQUAL INTERVALS)

$$y = y_0 + \frac{p}{1!} \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

where, $p = \frac{x - x_0}{h}$

h = Interval of differencing (Difference between consecutive values of x)

x_0 = Initial value of x

x = value of x for which $y = f(x)$ to be find out.

Newton-Gregory Backward Interpolation Formula: (EQUAL INTERVALS)

$$y = y_n + \frac{p}{1!} \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

where, $p = \frac{x - x_n}{h}$, h = Interval of differencing.

x_n = Final value of x

Note: From the given set of values if we want to find value of y at x which is nearer to the initial values then we use Newton's forward formula. If it is nearer to the end values, then use backward formula.

Q Find the value of y at $x=21$ from the table below.

x	20	23	26	29
y	0.3420	0.3907	0.4384	0.4848

Sol:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
$x_0 = 20$	<u>$0.3420 = y_0$</u>			
23	0.3907	0.0487	-1×10^{-3}	-3×10^{-4}
26	0.4384	0.0477	-1.3×10^{-3}	
29	0.4848	0.0464		

We use forward Interpolation formula, since we want to find y at $x=21$ which is nearer to the initial value

$$y = y_0 + \frac{P}{1!} \Delta y_0 + \frac{P(P-1)}{2!} \Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!} \Delta^3 y_0 \quad \text{--- (1)}$$

Here: $P = \frac{x - x_0}{h} = \frac{21 - 20}{3} = \frac{1}{3} \quad \therefore \text{(1) becomes,}$

$$\begin{aligned} y &= 0.3420 + \frac{1}{3} \cdot \frac{1}{1!} (0.0487) + \frac{1}{3} \left(\frac{1}{3} - 1 \right) \cdot \frac{1}{2!} (-1 \times 10^{-3}) \\ &\quad + \frac{1}{3} \left(\frac{1}{3} - 1 \right) \left(\frac{1}{3} - 2 \right) \cdot \frac{1}{3!} (-3 \times 10^{-4}) \\ &= \underline{\underline{0.35812}} \end{aligned}$$

Thus value of y at $x=21$ is 0.35812

OR $y(21) = \underline{\underline{0.35812}}$

NOTE: $y_0, \Delta y_0, \Delta^2 y_0, \Delta^3 y_0$ are the 1st values in the table which are marked.

Q Find the population ^{of City A} during the year 1900 from the given data

Year	1891	1901	1911	1921	1931	
Population in City A (in thousands)	46	66	81	93	101	

Sol:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1891	46	20			
1901	66	15	-5	2	
1911	81	12	-3	-1	-3
1921	93	8	-4		
1931	101				

$$y = y_0 + \frac{P}{1!} \Delta y_0 + \frac{P(P-1)}{2!} \Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!} \Delta^3 y_0 + \frac{P(P-1)(P-2)(P-3)}{4!} \Delta^4 y_0$$

①

$$P = \frac{x - x_0}{h} = \frac{1900 - 1891}{10} = \frac{9}{10} = 0.9 \quad \therefore \text{① becomes}$$

$$y = 46 + \frac{0.9}{1!} \times 20 + \frac{(0.9)(0.9-1)}{1.2} \times (-5) + \frac{(0.9)(0.9-1)(0.9-2)}{1.2 \cdot 3} \times 2 + \frac{(0.9)(0.9-1)(0.9-2)(0.9-3)}{1.2 \cdot 3 \cdot 4}$$

$$\neq 64.839$$

$$= \underline{\underline{64.2839}}$$

Thus population in City A during the year 1900 is 64 thousands.

- [Q] The following data gives the melting point of an alloy of lead and zinc, where 't' is the temperature in °C and p is the percentage of lead in the alloy.

P%	60	70	80	90
t	226	250	276	304

Find the melting point of the alloy containing 84% of lead, using Newton's interpolation formula.

Sol:

x (P%)	y (t)	∇y	$\nabla^2 y$	$\nabla^3 y$
60	226			
70	250	24	2	
80	276	26	2	0
90	304	28		

We use backward interpolation formula, since we want to find t for 84% which is nearer to the end values.

$$y = y_n + \frac{p}{1!} \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n \quad \text{--- (1)}$$

Here: $p = \frac{x - x_n}{h} = \frac{84 - 90}{10} = \frac{-6}{10} = -0.6$ $\therefore$ (1) becomes,

$$y = 304 + \frac{-0.6}{1!} \times 28 + \frac{(-0.6)(-0.6+1)}{2!} \times 2$$

$$= 304 + -16.8 + -0.24$$

$$= \underline{286.96} \quad \therefore \text{Thus melting point of alloy containing}$$

84% of lead is 286.96° C. NOTE: $y_n, \nabla y_n, \nabla^2 y_n$ are the last values in the table that is highlighted.

Q Use Newton's interpolation formula to estimate the increase in the population during the period 1946 to 1948. Given data is:

Year :	1911	1921	1931	1941	1951	1961
Population :	12	15	20	27	39	52
(in 1000's)						

Sol: We construct Newton's backward difference table

Year x	Pop. y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
1911	12	3	2	0	3	-10
1921	15	5	2	3	-7	
1931	20	7	5	-4		
1941	27	12	1			
1951	39	13				
1961	52					

To find the population in the year 1946

$$y = y_n + \frac{p}{1!} \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n + \frac{p(p+1)(p+2)(p+3)(p+4)}{5!} \nabla^5 y_n$$

$$p = \frac{x - x_n}{h} = \frac{1946 - 1961}{10} = -1.5$$

①

$$y_n = 52, \nabla y_n = 13, \nabla^2 y_n = 1, \nabla^3 y_n = -4, \nabla^4 y_n = -7, \nabla^5 y_n = -10$$

Substituting all these values in ①,

$$y = 32.578 \Rightarrow \text{Population in 1946 is } \underline{\underline{33 \text{ thousands}}}$$

To find the population in the year 1948

$$p = \frac{x - x_n}{h} = \frac{1948 - 1961}{10} = 1.3$$

Substituting the values in ①,

$$y = \underline{34.913} \Rightarrow \text{Population in 1948} = 35 \text{ thousands}$$

Thus required increase in population during 1946 to 1948 is 2 thousands.

Home work

Q The following are data from a steam table:

Temp. °C	: 140	150	160	170	180
Pressure kg/cm ²	: 3.685	4.854	6.302	8.076	10.225

Using Newton's formula, find the pressure of steam for a temperature 142°C.

KTU

Q Consider the data given in the following table.

x	0	0.5	1	1.5	2
f(x)	1	1.0513	1.1052	1.1618	1.2214

Estimate the value of $f(1.80)$ using Newton's backward interpolation formula.

Q1 Find the interpolating polynomial from the data given:

OR

Find the polynomial which represents the given data.

x :	0	1	2	3	4
y :	3	6	11	18	27

Sol: we construct forward difference table as follows: -

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	3	3	2	0
1	6	5	2	0
2	11	7	2	
3	18	9		
4	27			

Newton's forward interpolation formula is,

$$y = y_0 + \frac{p}{1!} \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \dots \quad \text{--- (1)}$$

Here, $p = \frac{x - x_0}{h} = \frac{x - 0}{1} = x \quad \therefore \text{(1) becomes,}$

$$y = 3 + \frac{x}{1!} \times 3 + \frac{x(x-1)}{2!} \times 2 + 0$$

$$= 3 + 3x + x^2 - x$$

$$= \underline{\underline{x^2 + 2x + 3}} \quad \text{is the required polynomial.}$$

NOTE: Here we ~~can~~ ^{may} construct backward difference table and use Newton's backward interpolation formula.

Home work

Q2 Find the cubic polynomial from the data given:

x :	0	2	4	6	8
$f(x) = y$:	7	13	43	145	367

Lagrange's Formula for Unequal Intervals :-

Let $y = f(x)$ be a function which takes the values $y_0, y_1, y_2, \dots, y_n$ when x assumes the values $x_0, x_1, \dots, x_n$ respectively. Here the values of x are not equally spaced. Then for finding a value of y corresponding to any value in the interval (x_0, x_n) we use Lagrange's Interpolation formula.

$$y = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} y_1 + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} y_n$$

[9] Using Lagrange's interpolation formula, find the value of y when $x=10$ from the following data.

x :	5	6	9	11
y :	12	13	14	16

Sol: Here : $x_0=5$ $y_0=12$
 $x_1=6$ $y_1=13$ Here : $x=10$
 $x_2=9$ $y_2=14$
 $x_3=11$ $y_3=16$

Lagrange's interpolation formula is,

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$\begin{aligned}
 y &= \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)} \times 12 + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)} \times 13 \\
 &+ \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)} \times 14 + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)} \times 16 \\
 &= \frac{4 \times 1 \times 12}{-1 \times -4 \times -6} + \frac{5 \times -1 \times 13}{-3 \times -5} + \frac{5 \times 4 \times -1 \times 14}{4 \times 3 \times -2} + \frac{5 \times 4 \times 16}{6 \times 5 \times 2} \\
 &= 12 + \frac{-13}{3} + \frac{35}{3} + \frac{32}{6} = \frac{12}{6} - \frac{26}{6} + \frac{70}{6} + \frac{32}{6} = \frac{14.66}{6}
 \end{aligned}$$

Q. Find the polynomial which satisfies the following data.

x:	0	2	3	6
y:	659	705	729	804

Sol: Since, x-values are not equally spaced, we use Lagrange's interpolation formula.

$$\begin{array}{ll}
 \text{Here: } x_0 = 0 & y_0 = 659 \\
 x_1 = 2 & y_1 = 705 \\
 x_2 = 3 & y_2 = 729 \\
 x_3 = 6 & y_3 = 804
 \end{array}$$

$$\begin{aligned}
 y &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \\
 &\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3 \\
 &= \frac{(x-2)(x-3)(x-6)}{-2 \times -3 \times -6} \times \frac{659}{-29} + \frac{(x-0)(x-3)(x-6)}{2 \times -1 \times -4} \times 705 \\
 &+ \frac{(x-0)(x-2)(x-6)}{3 \times 1 \times -3} \times 729 + \frac{(x-0)(x-2)(x-3)}{6 \times 4 \times 3} \times 804 = \frac{1}{72} [-x^3 + 29x^2 + 1602x + 47448]
 \end{aligned}$$

Home works :

- Q Find the value of $y=e^x$ when $x=0.38$ from the data given:

x	: 0	0.1	0.2	0.3	0.4
e^x	: 1	1.1052	1.2214	1.3499	1.4918

- Q From the given table, compute the value of $\sin 38^\circ$

x°	: 0	10	20	30	40
$\sin x^\circ$	: 0	0.17365	0.34202	0.5	0.64279

- Q Find $f(4)$ from the given data

x	: 0	2	3	6
$f(x)$	: -4	2	14	158

- Q From an experiment, we get the following values of a function $\phi(x)$:

x	: 1	2	-4
$\phi(x)$	: 3	-5	4

Express the function $\phi(x)$ approximately by a polynomial of degree 2

Inverse Lagranges formula

$$x = \frac{(y-y_1)(y-y_2)\dots(y-y_n)}{(y_0-y_1)(y_0-y_2)\dots(y_0-y_n)} x_0 + \frac{(y-y_0)(y-y_2)\dots(y-y_n)}{(y_1-y_0)(y_1-y_2)\dots(y_1-y_n)} x_1 \\ + \dots + \frac{(y-y_0)(y-y_1)\dots(y-y_{n-1})}{(y_n-y_0)(y_n-y_1)\dots(y_n-y_{n-1})} x_n$$

Q. Apply Inverse Lagranges formula to find value of x when $y=19$ from the data given:

x	0	1	2
y	0	1	20

Sol. Here: $x_0=0$ $y_0=0$
 $x_1=1$ $y_1=1$ we've to find 'x' if $y=19$
 $x_2=2$ $y_2=20$

Inverse Lagranges formula is,

$$x = \frac{(y-y_1)(y-y_2)}{(y_0-y_1)(y_0-y_2)} x_0 + \frac{(y-y_0)(y-y_2)}{(y_1-y_0)(y_1-y_2)} x_1 + \frac{(y-y_0)(y-y_1)}{(y_2-y_0)(y_2-y_1)} x_2$$

Substituting all the values, we get

$$x = \frac{(19-1)(19-20)}{(0-1)(0-20)} \times 0 + \frac{(19-0)(19-20)}{(1-0)(1-20)} \times 1 + \frac{(19-0)(19-1)}{(20-0)(20-1)} \times 2 \\ = \frac{19 \times -1}{-19} + \frac{19 \times 18}{20 \times 19} \times 2 = 1 + 1.8 = \underline{\underline{2.8}}$$

Thus, for $y=19$ value of $x = \underline{\underline{2.8}}$

Newton's Divided Difference Formula:

Let $f(x_0), f(x_1), f(x_2), \dots, f(x_n)$ be the values of $f(x)$ corresponding to the non-equally spaced arguments $x_0, x_1, x_2, \dots, x_n$. Then Newton's divided difference interpolation formula is given by,

$$y = f(x) = f(x_0) + (x - x_0) \Delta f(x_0) + (x - x_0)(x - x_1) \Delta^2 f(x_0) + (x - x_0)(x - x_1)(x - x_2) \Delta^3 f(x_0) + \dots$$

NOTE: The above formula can be used even when $x_0, x_1, \dots, x_n$ are equally spaced.

[Q] Using Newton's divided difference formula, evaluate $f(8)$ and $f(15)$ from the data given:

$x:$	4	5	7	10	11	13
$f(x):$	118	100	294	900	1210	2028

sol: Newton's divided difference formula is,

$$\begin{aligned} f(x) = y &= f(x_0) + (x - x_0) \Delta f(x_0) + (x - x_0)(x - x_1) \Delta^2 f(x_0) + \dots \\ &= f(4) + (x - 4) \Delta f(4) + (x - 4)(x - 5) \Delta^2 f(4) + \\ &\quad (x - 4)(x - 5)(x - 7) \Delta^3 f(4) + \dots \quad \text{--- (1)} \end{aligned}$$

Let us construct the divided difference table for getting the values of $\Delta f(x_0)$, $\Delta^2 f(x_0)$, $\dots$

Divided difference table:

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
4	48	$\frac{100-48}{5-4} = 52$	$\frac{97-52}{7-4} = 15$	$\frac{21-15}{10-4} = 1$	
5	100				0
7	294	$\frac{294-100}{7-5} = 97$	$\frac{202-97}{10-5} = 21$	$\frac{27-21}{11-5} = 1$	
10	900	$\frac{900-294}{10-7} = 202$	$\frac{310-202}{11-7} = 27$	$\frac{33-27}{13-7} = 1$	0
11	1210	$\frac{1210-900}{11-10} = 310$	$\frac{409-310}{13-10} = 33$		
13	2028	$\frac{2028-1210}{13-11} = 409$			

Here : $\Delta f(x_0) = 52$, $\Delta^2 f(x_0) = 15$, $\Delta^3 f(x_0) = 1$

ie $\Delta f(4) = 52$, $\Delta^2 f(4) = 15$, $\Delta^3 f(4) = 1$

Substituting all these values in (1) we get

$$f(x) = y = 48 + (x-4)52 + (x-4)(x-5)15 + (x-4)(x-5)(x-7) \cdot 1 \quad \text{--- (2)}$$

To find $f(8)$: Put $x=8$ in (2)

$$f(8) = y(8) = 48 + (8-4)52 + (8-4)(8-5)15 + (8-4)(8-5)(8-7) \cdot 1$$

$$\Rightarrow y(8) = \underline{\underline{448}} \quad \text{ie } \underline{\underline{f(8) = 448}}$$

To find $f(15)$: Put $x=15$ in (2)

$$f(15) = y(15) = 48 + (15-4)52 + (15-4)(15-5)15 + (15-4)(15-5)(15-7) = 3150$$

$$\text{ie } \underline{\underline{f(15) = 3150}}$$

Q Using Newton's formula find $f(x)$ from the following data:

x :	0	2	3	4	7	9
$f(x)$:	4	26	58	112	466	922

Sol: Since x -values are not equally spaced, we use Newton's divided difference formula.

Divided Difference Table

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
0	4	$\frac{26-4}{2-0} = 11$	$\frac{32-11}{3-0} = 7$	$\frac{11-7}{4-0} = 1$	0
2	26	$\frac{58-26}{3-2} = 32$	$\frac{54-32}{4-2} = 11$	$\frac{16-11}{7-2} = 1$	
3	58	$\frac{112-58}{4-3} = 54$	$\frac{118-54}{7-3} = 16$	$\frac{22-16}{9-3} = 1$	0
4	112	$\frac{466-112}{7-4} = 118$	$\frac{228-118}{9-4} = 22$		
7	466	$\frac{922-466}{9-7} = 228$			
9	922				

Newton's divided difference formula is,

$$f(x) = f(x_0) + (x-x_0) \Delta f(x_0) + (x-x_0)(x-x_1) \Delta^2 f(x_0) + (x-x_0)(x-x_1)(x-x_2) \Delta^3 f(x_0) + \dots$$

Here: $x_0 = 0$, $f(x_0) = 4$, $\Delta f(x_0) = 11$, $\Delta^2 f(x_0) = 7$, $\Delta^3 f(x_0) = 1$

$$\begin{aligned} \Rightarrow f(x) &= 4 + (x-0) 11 + (x-0)(x-2) 7 + (x-0)(x-2)(x-3) 1 \\ &= 4 + 11x + x(x-2) 7 + x(x-2)(x-3) \\ &= 4 + 11x + 7x^2 - 14x + (x^2 - 2x)(x-3), \text{ On simplification} \\ &= x^3 + 2x^2 + 3x + 4 // \end{aligned}$$

Q Find value of $f(10)$ by Newton Divided difference method.

x	5	6	9	11
$f(x)=y$	12	13	14	16

Sol: Divided difference Table:

	x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
$x_0 \rightarrow$	5	12	$\frac{13-12}{6-5} = 1$	$\frac{\frac{1}{3}-1}{9-5} = -\frac{1}{6}$	$\frac{\frac{2}{5}-\frac{1}{6}}{11-5} = \frac{1}{20}$
$x_1 \rightarrow$	6	13	$\frac{14-13}{9-6} = \frac{1}{3}$		
$x_2 \rightarrow$	9	14	$\frac{16-14}{11-9} = 1$	$\frac{1-\frac{1}{3}}{11-6} = \frac{2}{15}$	
$x_3 \rightarrow$	11	16			

Newtons divided difference formula is,

$$y = f(x) = f(x_0) + (x-x_0)\Delta f(x_0) + (x-x_0)(x-x_1)\Delta^2 f(x_0) + \dots \quad (1)$$

Here we need to find $f(10)$. So, we put $x = 10$,

$$x_0 = 5, \Delta f(x_0) = 1, \Delta^2 f(x_0) = -\frac{1}{6}, \Delta^3 f(x_0) = \frac{1}{20}$$

$$f(x_0) = 12$$

Substituting all values in (1),

$$y = f(10) = 12 + (10-5) \cdot 1 + (10-5)(10-6) \left(-\frac{1}{6}\right) + (10-5)(10-6)(10-9) \times \frac{1}{20}$$

$$= 12 + 5 + \left(-\frac{20}{6}\right) + 1$$

$$= 18 - \frac{20}{6} = \frac{44}{3} = \underline{\underline{14.66}}$$

Numerical Integration:

The process of computing the value of a definite integral $\int_a^b y dx$ from a set of numerical values of the integrand y is called numerical integration.

Trapezoidal Rule:

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

Simpson's one-third rule

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$$

NOTE: while using Simpson's $\frac{1}{3}$ rd rule, given interval of integration must be divided into an even number of sub-intervals.

Q Use trapezoidal rule to evaluate $\int_0^1 x^3 dx$ considering five sub-intervals.

Sol: Dividing the interval $(0,1)$ into five equal parts, we get, step size h as, $h = \frac{1-0}{5} = 0.2$. Thus, x values are,

$$x_0 = 0, x_1 = 0.2, x_2 = 0.4, x_3 = 0.6, x_4 = 0.8, x_5 = 1$$

Corresponding to each of these x values, we find values of $y = x^3$

Thus,

$$x_0 = 0 \rightarrow y_0 = x^3 = 0$$

$$x_1 = 0.2 \rightarrow y_1 = (0.2)^3 = 0.008$$

$$x_2 = 0.4 \rightarrow y_2 = (0.4)^3 = 0.064$$

$$x_3 = 0.6 \rightarrow y_3 = (0.6)^3 = 0.216$$

$$x_4 = 0.8 \rightarrow y_4 = (0.8)^3 = 0.512$$

$$x_5 = 1 \rightarrow y_5 = (1)^3 = 1$$

By the Trapezoidal rule, we have

$$\int_{x_0}^{x_n} x^3 dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

$$\Rightarrow \int_0^1 x^3 dx = \frac{0.2}{2} [(y_0 + y_5) + 2(y_1 + y_2 + y_3 + y_4)]$$

$$= \frac{0.2}{2} [(0 + 1) + 2(0.008 + 0.064 + 0.216 + 0.512)]$$

$$= \underline{\underline{0.26}}$$

[Q] Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Simpson's $\frac{1}{3}$ rule

taking $h = \frac{1}{4}$.

Sol: Given that $h = \frac{1}{4}$. So interval $(0, 1)$ must be divided with step size $\frac{1}{4}$. ie Thus x values are $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$. Corresponding to each of these x -values we find values of $y = \frac{1}{1+x^2}$.

Thus,

$$x_0 = 0 \rightarrow y_0 = \frac{1}{1+0} = 1$$

$$\boxed{y = \frac{1}{1+x^2}} = f(x)$$

$$x_1 = \frac{1}{4} \rightarrow y_1 = \frac{1}{1+(\frac{1}{4})^2} = 0.9412$$

$$x_2 = \frac{1}{2} \rightarrow y_2 = \frac{1}{1+(\frac{1}{2})^2} = 0.8$$

$$x_3 = \frac{3}{4} \rightarrow y_3 = \frac{1}{1+(\frac{3}{4})^2} = 0.64$$

$$x_4 = 1 \rightarrow y_4 = \frac{1}{1+1^2} = 0.5$$

By Simpson's $\frac{1}{3}$ rd rule,

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2}) \right]$$

Here,

$$\begin{aligned} \int_0^1 \frac{1}{1+x^2} dx &= \frac{1/4}{3} \left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2) \right] \\ &= \frac{1}{12} \left[(1 + 0.5) + 4(0.9412 + 0.64) + 2(0.8) \right] \\ &= \underline{\underline{0.7854}} \end{aligned}$$

KTU

Q Evaluate $\int_0^1 e^{-x^2/2} dx$ using (i) trapezoidal rule.

(ii) Simpson's $\frac{1}{3}$ rd rule

Sol. Trapezoidal rule Let $h = 0.2$. Then, from the interval $(0, 1)$, we get

x values as : $0, 0.2, 0.4, 0.6, 0.8, 1$.

Corresponding to each of these x -values, we find y values of $y = e^{-x^2/2}$

Thus,

$$x_0 = 0 \rightarrow y_0 = e^{-0} = 1$$

$$\boxed{y = e^{-x^2/2}} = f(x)$$

$$x_1 = 0.2 \rightarrow y_1 = e^{-\frac{(0.2)^2}{2}} = 0.98019$$

$$x_2 = 0.4 \rightarrow y_2 = e^{-\frac{(0.4)^2}{2}} = 0.92311$$

$$x_3 = 0.6 \rightarrow y_3 = e^{-\frac{(0.6)^2}{2}} = 0.83527$$

$$x_4 = 0.8 \rightarrow y_4 = e^{-\frac{(0.8)^2}{2}} = 0.72614$$

$$x_5 = 1 \rightarrow y_5 = e^{-\frac{1^2}{2}} = 0.60653$$

Trapezoidal rule is,

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

Here,

$$\begin{aligned} \int_0^1 e^{-x^2/2} dx &= \frac{0.2}{2} [(y_0 + y_5) + 2(y_1 + y_2 + y_3 + y_4)] \\ &= \frac{0.2}{2} [(1 + 0.60653) + 2(0.98019 + 0.92311 \\ &\quad + 0.83527 + 0.72614)] \\ &= 0.8535999 \end{aligned}$$

Simpson's $\frac{1}{3}$ rd rule

While using Simpson's $\frac{1}{3}$ rd rule the given interval (0,1) must be divided into an even number of sub-intervals.

In trapezoidal rule we applied just now,
 we had $x_0, x_1, x_2, x_3, x_4, x_5$ ~~and~~, thus
 $\underbrace{\quad}_1 \quad \underbrace{\quad}_2 \quad \underbrace{\quad}_3 \quad \underbrace{\quad}_4 \quad \underbrace{\quad}_5$
 5 no. of subintervals. So previous data set
 we can't make use for Simpson's rule.

Let us divide the interval $(0,1)$ into 8 equal
 parts. we get step size h as, $h = \frac{1-0}{8} = 1/8 = 0.125$
 Thus, we have:

$$x_0 = 0 \rightarrow y_0 = e^{-0} = 1$$

$$y = e^{-x/2}$$

$$x_1 = 0.125 \rightarrow y_1 = e^{-(0.125)/2} = 0.992217$$

$$x_2 = 0.250 \rightarrow y_2 = 0.969233$$

$$x_3 = 0.375 \rightarrow y_3 = 0.932102$$

$$x_4 = 0.5 \rightarrow y_4 = 0.88249$$

$$x_5 = 0.625 \rightarrow y_5 = 0.822577$$

$$x_6 = 0.750 \rightarrow y_6 = 0.754839$$

$$x_7 = 0.875 \rightarrow y_7 = 0.6819408$$

$$x_8 = 1 \rightarrow y_8 = 0.60653$$

$$\text{Simpson's } \frac{1}{3}^{\text{rd}} \text{ rule: } \int_{x_0}^{x_n} y(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + \dots + y_{n-1}) + 2(y_2 + \dots + y_{n-2})]$$

$$\Rightarrow \int_0^1 e^{-x/2} dx = \frac{0.125}{3} [(y_0 + y_8) + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)]$$

$$= 0.855626 //$$

Q The speed of a moving particle was measured at different points of time. The time t when the 1st measurement was recorded is taken as $t=0$. Subsequent speeds at different times are as shown below.

Time (t): 0 10 20 30 40 50 60
(in seconds)

Velocity (v): 35 39 44 50 56 43 40
(in m/s) v_0 v_1 v_2 v_3 v_4 v_5 v_6

Evaluate the distance travelled by the particle in 60 sec

Sol: Total distance travelled in 60 seconds = $\int_0^{60} v \, dt$

In the data given, there are 6 sub-intervals. (even).
So we can use Simpson's $\frac{1}{3}$ rd rule to evaluate $\int_0^{60} v \, dt$.

$$\Rightarrow \int_0^{60} v \, dt = \frac{h}{3} [(v_0 + v_6) + 4(v_1 + v_3 + \dots) + 2(v_2 + v_4 + \dots)]$$

[Here], $= \frac{10}{3} [(v_0 + v_6) + 4(v_1 + v_3 + v_5) + 2(v_2 + v_4)]$

$$= \frac{10}{3} [(35 + 40) + 4(39 + 50 + 43) + 2(44 + 56)]$$

$$= \frac{10}{3} [75 + (4 \times 132) + 2 \times 100]$$

$$= 2676.67$$

$\therefore$ Total distance travelled in 60 seconds = 2676.67 m

- Q Evaluate $\int_4^{5.2} \log_e x \, dx$ using (i) Trapezoidal rule
 (ii) Simpson's rule and
 (iii) compare the answers by actual integration

Sol. Given interval is $(4, 5.2)$. Let $h = 0.2$. Then values of x are : 4, 4.2, 4.4, 4.6, 4.8, 5, 5.2. Corresponding to each of these x -values, we find values for $y = \log_e x$

$$x_0 = 4 \rightarrow y_0 = \log_e 4 = 1.3863$$

$$y = \log_e x$$

$$x_1 = 4.2 \rightarrow y_1 = \log_e 4.2 = 1.4351$$

$$x_2 = 4.4 \rightarrow y_2 = \log_e 4.4 = 1.4816$$

$$x_3 = 4.6 \rightarrow y_3 = \log_e 4.6 = 1.5261$$

$$x_4 = 4.8 \rightarrow y_4 = \log_e 4.8 = 1.5686$$

$$x_5 = 5 \rightarrow y_5 = \log_e 5 = 1.6094$$

$$x_6 = 5.2 \rightarrow y_6 = \log_e 5.2 = 1.6487$$

NOTE

Natural logarithm	$\log_e x = \ln x$
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Common logarithm	$\log_{10} x = \log x$
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(i) Trapezoidal rule

$$\int_{x_0}^{x_n} f(x) \, dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

Here, $f(x) = \log_e x$, $h = 0.2$, $x_0 = 4$, $x_6 = 5.2$, $(y_0, \dots, y_6)$ (see table)

$$\begin{aligned} \int_4^{5.2} \log_e x \, dx &= \frac{0.2}{2} [(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)] \\ &= \underline{\underline{1.82766}} \end{aligned}$$

(ii) Simpson's $\frac{1}{3}$ rd rule : Since the number of sub-intervals are 6 (even) we can use the ^{above} same data

to apply Simpson's rule.

Simpson's $\frac{1}{3}$ rd rule:

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots)]$$

Here: $x_0 = 4$, $x_n = 5.2$, $f(x) = \log_e x$, $h = 0.2$. Thus,

$$\begin{aligned} \int_4^{5.2} \log_e x &= \frac{0.2}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)] \\ &= \underline{\underline{1.82785}} \end{aligned}$$

(iii) Comparison of answers with actual integration.

$$\int_4^{5.2} \log_e x = \left[x \log_e x - x \right]_4^{5.2}$$

$$\boxed{\int \log x dx = x \log x - x}$$

$$= (5.2 \log_e 5.2 - 5.2) - (4 \log_e 4 - 4) = \underline{\underline{1.82782}}$$

Here, answer got by Simpson's $\frac{1}{3}$ rd rule is more closer to exact value than Trapezoidal rule answer.

Homework

Q The following table gives the values of a function $f(x)$ for certain values of x .

x	0	0.25	0.50	0.75	1
$f(x)$	1	0.9412	0.8	0.64	0.5

Evaluate $\int_0^1 f(x) dx$ using Trapezoidal rule.

Q Evaluate $\int_1^2 \frac{dx}{x}$ using Simpson's rule and

Trapezoidal rule by taking $h = 0.25$. Also, compare the answer with exact value by actual integration.

Q By dividing the range into 10 equal parts, evaluate $\int_0^{\pi} \sin x \, dx$ using Simpson's rule and verify the result by actual integration.

Sol: Dividing the range $(0, \pi)$ into 10 equal parts $\Rightarrow h = \frac{\pi - 0}{10}$

Thus, x values are,

$$0, \frac{\pi}{10}, \frac{2\pi}{10}, \frac{3\pi}{10}, \frac{4\pi}{10}, \frac{5\pi}{10}, \frac{6\pi}{10}, \frac{7\pi}{10}, \frac{8\pi}{10}, \frac{9\pi}{10}, \frac{10\pi}{10} = \pi$$

Corresponding to each of these x -values, we find f values of $y = \sin x$.

$$x_0 = 0 \rightarrow y_0 = \sin 0 = 0$$

$$x_1 = \pi/10 \rightarrow y_1 = \sin \pi/10 = 0.3090$$

$$x_2 = 2\pi/10 \rightarrow y_2 = \sin 2\pi/10 = 0.5878$$

$$x_3 = 3\pi/10 \rightarrow y_3 = \sin 3\pi/10 = 0.809$$

$$x_4 = 4\pi/10 \rightarrow y_4 = \sin 4\pi/10 = 0.9511$$

$$x_5 = 5\pi/10 \rightarrow y_5 = \sin 5\pi/10 = 1$$

$$x_6 = 6\pi/10 \rightarrow y_6 = \sin 6\pi/10 = 0.9511$$

$$x_7 = 7\pi/10 \rightarrow y_7 = \sin 7\pi/10 = 0.809$$

$$x_8 = 8\pi/10 \rightarrow y_8 = \sin 8\pi/10 = 0.5878$$

$$x_9 = 9\pi/10 \rightarrow y_9 = \sin 9\pi/10 = 0.3090$$

$$x_{10} = \frac{10\pi}{10} \rightarrow y_{10} = \sin \pi = 0$$

$$\text{Simpson's } \frac{1}{3}^{\text{rd}} \text{ rule: } \int_{x_0}^{x_n} f(x) \, dx = \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots) \right]$$

$$\int_0^{\pi} \sin x \, dx = \frac{\pi/10}{3} \left[(y_0 + y_{10}) + 4(y_1 + y_3 + y_5 + y_7 + y_9) + 2(y_2 + y_4 + y_6 + y_8) \right] = 2.00091$$

$$\boxed{y = \sin x}$$

By actual integration

$$\begin{aligned} \int_0^{\pi} \sin x \, dx &= [-\cos x]_0^{\pi} \\ &= [-\cos \pi - (-\cos 0)] \\ &= [-(-1) - (-1)] \\ &= \underline{\underline{2}} \end{aligned}$$

- [Q] A river is 80 meters wide. The depth 'd' in meters at a distance 'x' meters from one bank is given by the following table. Calculate the area of cross-section of the river using Simpson's rule.

	x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
x_m	0	10	20	30	40	50	60	70	80
d_m	0	4	7	9	12	15	14	8	3

Sol. Simpson's rule: $\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots)]$

Here, $h = 10$.

$$\text{Area} = \int_0^{80} d(x) dx = \frac{10}{3} [(y_0 + y_8) + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)]$$

$$= \underline{\underline{710}} \text{ m}^2$$

- [Q] A curve passes through the points (x_i, y_i) , $i = 0, 1, \dots, 6$. Find the volume of solid of revolution about the x-axis.

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	1	1.5	2	2.5	3	3.5	4
y	2	2.4	2.7	2.8	3	2.6	2.1

Sol. We know, volume of solid of revolution about the x-axis $= \int_{x_0}^{x_n} \pi y^2 dx$

From the table, we see that $h = 0.5$.

By Simpson's $\frac{1}{3}$ rule,

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots)]$$

NOTE:

volume of solid of revolution about y-axis $= \int_{y_0}^{y_n} \pi x^2 dy$

Here,

$$\text{Volume} = \int_{x_0}^{x_n} \pi y^2 dx$$

$$= \int_1^4 \pi y^2 dx$$

$$= \pi \int_1^4 y^2 dx$$

$$= \pi \left[\frac{h}{3} \left\{ (y_0^2 + y_6^2) + 4(y_1^2 + y_3^2 + y_5^2) + 2(y_2^2 + y_4^2) \right\} \right]$$

$$= \pi \left[\frac{0.5}{3} \left\{ (2^2 + (2.6)^2) + 4(2.4^2 + 2.8^2 + 2.6^2) + 2(2.7^2 + 3^2) \right\} \right]$$

$$= \underline{\underline{64.088 \text{ units}}}$$

Note:

In the table we have only y-values. Here $f(x) = y^2$. So we need, $y_0^2, y_1^2, \dots$ corresponding to each $x_0, x_1, \dots$

- Q The acceleration of a moving particle is measured in every 5 seconds and is tabulated as below. Find velocity of the particle using (i) Trapezoidal rule (ii) Simpson's $\frac{1}{3}$ rd rule.

t(sec)	0	5	10	15	20	25	30	35	40
a (m/sec ²)	40	45.25	48.5	51.25	54.35	59.48	61.5	64.3	68.7
	a_0	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8

Sol.

Given: Acceleration, $a = \frac{dv}{dt}$

$$\therefore \text{velocity, } v = \int_{t_0}^{t_n} a dt = \int_0^{40} a dt$$

Given : $h = 5$

(i) Trapezoidal rule:

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

Here:

$$\begin{aligned} V &= \int_{t_0}^{t_n} a dt = \int_0^{40} a dt = \frac{5}{2} [(a_0 + a_8) + 2(a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7)] \\ &= \frac{5}{2} [(40 + 68.7) + 2(45.25 + 48.5 + 51.25 + 54.35 + 59.48 + 61.5 + 64.3)] \\ &= \underline{\underline{2194.9 \text{ m/s}}} \end{aligned}$$

(ii) Simpson's $\frac{1}{3}$ rd rule:

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots)]$$

Here:

$$\begin{aligned} V &= \int_{t_0}^{t_n} a dt = \int_0^4 a dt = \frac{5}{3} [(a_0 + a_8) + 4(a_1 + a_3 + a_5 + a_7) + 2(a_2 + a_4 + a_6)] \\ &= \frac{5}{3} [(40 + 68.7) + 4(45.25 + 51.25 + 59.48 + 64.3) + 2(48.5 + 54.35 + 61.5)] \\ &= \underline{\underline{2197.5 \text{ m/s}}} \end{aligned}$$

NOTE: Here we could apply Simpson's $\frac{1}{3}$ rd rule as the number of subintervals given in the table is even.

Home work:

- Q Evaluate $\int_0^1 x e^x dx$ using Simpson's rule with 8 sub intervals. Compare the result with actual value.
(Ans: 8.3897, 8.389056)